

DARTHI Y J

AI SOFTWARE ENGINEER | PYTHON BACKEND DEVELOPER

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PROFESSIONAL SUMMARY

AI Developer with strong expertise in Python and hands-on experience building Generative AI solutions using LLMs, RAG architectures, LangChain, LangGraph, and multi-agent orchestration (MCP, A2A protocols). Experienced in traditional machine learning (classification, clustering, regression), SQL-based data modeling, and ETL pipeline development. Proven ability to design scalable REST APIs using FastAPI, integrate vector databases, and deploy AI systems on cloud platforms including Microsoft Azure with CI/CD pipelines.

TECHNICAL SKILLS

Programming Languages : Python (Advanced), Java (Advanced), C (Advanced)

Backend and API Development : FastAPI, Flask, REST API, Pydantic, API Load Testing

AI / LLM Engineering : Large Language Models (LLMs), Hugging Face, PyTorch, LoRA, RAG, MCP and A2A Protocols

Generative AI and NLP : Text parsing, regex extraction, LangChain, LangGraph, Prompt Engineering

Machine Learning: Classification, Clustering, Regression, Predictive Modeling

Data Engineering : ETL Pipelines, Schema-Aware Processing, SCD Type 1 and Type 2, Incremental Loads

Databases: MySQL, PostgreSQL, SQLite, Redshift, MariaDB

Vector Databases : FAISS, Pinecone, ChromaDB

Cloud and DevOps : Docker, Google Cloud Run, Microsoft Azure, Render, CI/CD Pipelines, GitHub Actions

APIs and Integrations : OpenAI API, AWS Bedrock (Claude), Groq API, Google Gemini API

Web: HTML, CSS, JavaScript, React.js, JSP, Servlet, Bootstrap

Tools: VS Code, Eclipse EE, GitHub / Git, GitLab, Cursor, Postman, JMeter

WORK EXPERIENCE

AI SOFTWARE ENGINEER

AINXG fitness private limited ,Coimbatore | Nov 2025 – Present

AI-BASED CALORIE AND WORKOUT RECOMMENDATION SYSTEM

- Developed AI-driven fitness features using LLMs and RAG, including meal plan generation, vision-based food identification, and workout recommendations.
- Designed RAG-based meal planning pipelines with structured JSON outputs, calorie validation, and robust fallback logic.
- Built and benchmarked vision-enabled LLM workflows for accurate food recognition.
- Contributed to ML-based workout recommendation systems using Hugging Face models.

Tech Stack : Python, FastAPI, Hugging Face Transformers, PyTorch, Vision LLMs, LoRA, PEFT, Docker

AI PYTHON DEVELOPER - Cygnussoftwares , Vadavalli

CANDOR DATA PLATFORM | Jul 2025 – Oct 2025

- Worked on an AI-powered data engineering platform converting natural language prompts into Python ETL scripts and SQL pipelines.
- Integrated LLMs via AWS Bedrock to extract structured JSON and drive template-based ETL generation.
- Implemented schema-aware ETL pipelines with transformations, incremental loads, and SCD Type 1 and Type 2 logic.
- Contributed to ETL debugging tools, AI chat features, and payment workflow integration.

Tech Stack : Python, TypeScript, FastAPI, React, PostgreSQL, SQLite, Redshift, OpenAI API, ETL Pipelines, SCD Type 1 and Type 2, Schema-aware AI scripting

SALES INQUIRY AUTOMATION SYSTEM | May 2025 – Jun 2025

- Built an AI-driven email inquiry automation system for classification and response generation.
- Developed backend APIs using FastAPI and integrated frontend components using React.
- Automated email ingestion, classification, escalation, and response workflows.

Tech Stack : Python, FastAPI, React.js, SQLite, AI/ML Models, Task Scheduling, Batched Processing

EDUCATION

B.E. – Computer Science and Engineering | 2021 – 2025

RVS College of Engineering and Technology, Coimbatore

CGPA: 8.1

PROJECTS

- Ultra AI Agent – Modular AI assistant platform with prompt-to-JSON automation (Python, Flask, OpenAI API).
- AI Digital ADA Checker and Fixer – Automated ADA/WCAG accessibility audit and auto-fix system (Python, Selenium, BeautifulSoup).
- Parking Efficiency Management – Real-time parking CRUD system (JSP, Java EE, MySQL).
- Online Art Gallery – Responsive artwork showcase site (HTML, CSS, Bootstrap).
- Remote Exploitation Simulation – Windows 7 vulnerability analysis and exploitation.

CERTIFICATIONS AND INTERNSHIP

- Certified Internship in Cybersecurity.
- Python Development Internship – Data handling and Flask backend development.
- Java In-Plant Training – Core Java, JDBC, and database integration.
- NPTEL Certification – Java Programming.
- NPTEL Certification – C Programming.
- Advanced C Programming.
- Data Structures in C.
- Professional Workshops – Web Development and Data Science.

SOFT SKILLS

Problem Solving | Communication | Leadership | Teamwork | Time Management | Analytical Thinking